Haefliger's Theorem and Novikov's Theorem in Codimension 1

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1 Foliations and holonomy

Definition 1. Let M be a smooth manifold of dimension n. A *foliation atlas of codimension* 1 on M is an atlas $(\varphi_i : U_i \longrightarrow \mathbb{R}^n = \mathbb{R}^{n-1} \times \mathbb{R})_{i \in I}$ whose change-of-charts diffeomorphisms are locally of the form

$$\varphi_i \circ \varphi_j^{-1}(x, y) = \big(g_{ij}(x, y), h_{ij}(y)\big)$$

with respect to the decomposition $\mathbb{R}^n = \mathbb{R}^{n-1} \times \mathbb{R}$. The charts are *foliation charts*. The connected components of the submanifolds $\varphi_i^{-1}(\mathbb{R}^{n-1} \times \{y\})$ are the *plaques*. The plaques globally amalgamate into *leaves*, which are smooth manifolds of dimension $n-1$ injectively immersed in M. A *foliation of codimension* 1 is a maximal foliation atlas of codimension 1.

Definition 2. A codimension 1 foliation $\mathcal{F}$ can be represented by a *Haefliger cocycle*: an open cover (U_i) of M, submersions $s_i : U_i \longrightarrow \mathbb{R}$, and diffeomorphisms $\gamma_{ij} : s_j(U_i \cap U_j) \longrightarrow s_i(U_i \cap U_j)$ such that

$$\gamma_{ij} \circ s_j\big|_{U_i \cap U_j} = s_i\big|_{U_i \cap U_j}.$$

The diffeomorphisms γ_{ij} satisfy the cocycle condition

$$\gamma_{ij} \circ \gamma_{jk} = \gamma_{ik}$$

wherever both sides are defined. A codimension 1 foliation is transversely orientable if and only if it can be represented by a Haefliger cocycle with $d\gamma_{ij} > 0$.

Let $f : N \to M$ be a smooth map and let $\mathcal{F}$ be a codimension 1 foliation on M. If f is transversal to $\mathcal{F}$, meaning

$$(df)_x(T_x N) + T_{f(x)}(\mathcal{F}) = T_{f(x)} M$$

for every $x \in N$, then f pulls $\mathcal{F}$ back to a foliation $f^*(\mathcal{F})$ on N. If (U_i, s_i, γ_{ij}) represents $\mathcal{F}$, then $f^*(\mathcal{F})$ is represented on $f^{-1}(U_i)$ by the submersions $s_i \circ f$.

Definition 3. Let $(M, \mathcal{F})$ be a foliated manifold of codimension 1. A *transversal section* at $x \in M$ is a 1-dimensional submanifold of M through x which is transversal to the leaves. If α is a path in a leaf L from $x \in T$ to $y \in S$, where T and S are transversal sections, then α determines the holonomy germ $\mathrm{hol}_{S,T}(\alpha) : (T, x) \longrightarrow (S, y)$. For loops in L based at x, and for a fixed transversal section T at x, this gives a group homomorphism $\mathrm{hol}_T : \pi_1(L, x) \longrightarrow \mathrm{Diff}_x(T)$. After identifying (T, x) with $(\mathbb{R}, 0)$, this is written

$$\mathrm{hol} : \pi_1(L, x) \longrightarrow \mathrm{Diff}_0(\mathbb{R}),$$

uniquely up to conjugation in $\mathrm{Diff}_0(\mathbb{R})$. Its image is the holonomy group $\mathrm{Hol}(L, x)$.

A codimension 1 foliation is given by a nowhere vanishing integrable 1-form ω if and only if it is transversely orientable. In this case

$$T(\mathcal{F}) = \ker(\omega), \qquad d\omega \wedge \omega = 0.$$

A choice of transverse orientation is an orientation of the normal bundle $N(\mathcal{F}) = T(M)/T(\mathcal{F})$; with this choice, holonomy germs preserve the orientation of transversal sections, so $\mathrm{Hol}(L, x)$ is represented in $\mathrm{Diff}_0^+(\mathbb{R})$.

Definition 4. For a codimension 1 foliation, a holonomy germ of a loop is a germ of a diffeomorphism $g : (\mathbb{R}, 0) \longrightarrow (\mathbb{R}, 0)$. Such a germ is called *one-sided* if it restricts to the identity germ either on the negative half-germ $((-\infty, 0], 0)$ or on the positive half-germ $([0, \infty), 0)$. The foliation is *without non-trivial one-sided holonomy* if every one-sided holonomy germ is the identity germ. Analytic codimension 1 foliations have no non-trivial one-sided holonomy.

2 Morse functions into codimension 1 foliations; the disk index

Definition 5. Let $(N, \mathcal{F})$ be a codimension 1 foliated manifold and let X be a manifold. A smooth map $f : X \longrightarrow N$ is a *Morse function from X to $(N, \mathcal{F})$* if there exists a Haefliger cocycle (W_i, s_i, γ_{ij}) representing $\mathcal{F}$ such that every function

$$s_i \circ f\big|_{f^{-1}(W_i)} : f^{-1}(W_i) \longrightarrow \mathbb{R}$$

is a Morse function. A point $p \in X$ is a *singularity* of f if it is a singularity of one of these functions $s_i \circ f$.

Since $s_i = \gamma_{ij} \circ s_j$ on overlaps and γ_{ij} is a local diffeomorphism, the Morse condition and the Morse index of a singularity are independent of the chosen Haefliger chart, after a transverse orientation has been fixed.

Let D be a disk capping a loop transverse to $\mathcal{F}$. After perturbing the capping map rel boundary, we may assume that the resulting map $K : D \longrightarrow M$ is a Morse function from D to $(M, \mathcal{F})$, agrees with the original capping map near ∂D, and has critical values on distinct leaves. The pull-back $K^*(\mathcal{F})$ is then a singular 1-dimensional foliation on D. Its only singularities are the two Morse types:

- *centres* (elliptic singularities), of Morse index 0 or 2, locally given by the level sets of $x^2 + y^2$;
- *saddles*, of Morse index 1, locally given by the level sets of $x^2 - y^2$.

For this singular foliation of the disk, the number of centres is one bigger than the number of saddles:

$$\#\{\text{centres}\} - \#\{\text{saddles}\} = 1.$$

3 Haefliger's theorem

Theorem 1 (Haefliger)**.** *There are no analytic codimension 1 foliations on S^3.*

Theorem 2 (Haefliger)**.** *Let $(M, \mathcal{F})$ be a codimension 1 foliated manifold without non-trivial one-sided holonomy. Then every loop in M transverse to $\mathcal{F}$ represents an element of $\pi_1(M)$ of infinite order.*

Proof sketch. Suppose that a loop $\alpha : S^1 \to M$ transverse to $\mathcal{F}$ has finite order in $\pi_1(M)$. After replacing α by a power and smoothing the concatenation, we may assume $[\alpha] = 1$. Choose an extension $H : D \longrightarrow M$, $H|_{\partial D} = \alpha$. Since α is transverse to $\mathcal{F}$, the map H is transverse to $\mathcal{F}$ near ∂D. Perturb H, fixed near ∂D, to a Morse function $K : D \longrightarrow (M, \mathcal{F})$ whose critical values lie on distinct leaves. The pull-back $K^*(\mathcal{F})$ is a singular foliation of D, transverse to ∂D, with centres and saddles satisfying

$$\#\{\text{centres}\} - \#\{\text{saddles}\} = 1.$$

Choose a centre p. Around p there is a family of concentric circular leaves of $K^*(\mathcal{F})$. Let Γ be the boundary of the union of all such concentric circles. If Γ contains no singularity, then Γ is the last concentric circle around p. The leaves outside Γ spiral around it, so $K(\Gamma)$ has holonomy which is trivial on the side facing the concentric circles and non-trivial on the other side. This is a non-trivial one-sided holonomy germ, contradicting the hypothesis.

It remains to consider the case in which every such boundary curve contains a saddle. Since there is one more centre than saddle, two centres have boundary curves Γ_1 and Γ_2 meeting the same saddle. Locally near the saddle, only two configurations can occur: either $\Gamma_1 \cap \Gamma_2$ is exactly the saddle, or one boundary curve is contained in the other. In the first case set $\Gamma = \Gamma_1 \cup \Gamma_2$; in the second case set $\Gamma = (\Gamma_2 - \Gamma_1) \cup \{\text{the saddle}\}$. Each Γ_i has trivial holonomy on the side of its concentric circles. Because non-trivial one-sided holonomy is excluded, each $K(\Gamma_i)$ has trivial holonomy, and hence the corresponding composite curve $K(\Gamma)$ has trivial holonomy. Therefore there are concentric circles on the outside of Γ. Replacing the inside of Γ by this family of concentric circles removes singularities. Induction on the number of singularities eventually reaches the first case, again producing non-trivial one-sided holonomy. Thus no transverse loop can have finite order in $\pi_1(M)$. □

For example, the Kronecker foliation of the torus T^2 is induced by the submersion $s : \mathbb{R}^2 \to \mathbb{R}$, $s(x, y) = x - ay$, with a irrational. Its leaves are diffeomorphic to $\mathbb{R}$, hence have trivial holonomy, and they are dense in T^2. A circle transverse to this foliation represents an element of infinite order in $\pi_1(T^2) \cong \mathbb{Z}^2$.

4 Novikov's theorem

Theorem 3 (Novikov)**.** *Let $\mathcal{F}$ be a codimension 1 transversely oriented foliation of a compact manifold M of dimension 3 with finite fundamental group. Then $\mathcal{F}$ has a compact leaf.*

Definition 6. Let L_0 be a leaf of $(M, \mathcal{F})$. A closed curve $\alpha_0 : S^1 \longrightarrow L_0$ is a *vanishing cycle* if α_0 is not contractible in L_0, but for some $\varepsilon > 0$ it extends to a smooth family $\alpha : S^1 \times [0, \varepsilon) \longrightarrow M$, $\alpha_t = \alpha(-, t)$, such that each α_t lies in a leaf L_t, each α_t is contractible in L_t for $0 < t < \varepsilon$, and each segment $\alpha(s, -)$ is transverse to $\mathcal{F}$. The vanishing cycle is *positive* or *negative* according as these transverse segments are positive or negative with respect to the transverse orientation.

Equivalently, after choosing transversal sections smoothly along α_0, the family can be taken to be the holonomy extension

$$B : [0,1] \times [0,\varepsilon) \longrightarrow M, \qquad B(s,t) = \mathrm{hol}(\alpha_0|_{[0,s]})(\tau_{\alpha_0(0)}(t)).$$

Definition 7. A vanishing cycle α_0 is *simple* if, for sufficiently small $t > 0$, the lift of α_t to the universal covering space $\widetilde{L}_t$ of L_t is a simple closed curve.

Our reference first shows that every positive vanishing cycle leads, in arbitrarily nearby leaves, to a simple positive vanishing cycle. Thus, in proving compactness of a leaf, we may assume that the vanishing cycle is simple.

Proof sketch of Novikov's theorem. Step 1: a vanishing cycle exists. Since M is compact and $\mathcal{F}$ is transversely oriented, choose a loop α transverse to $\mathcal{F}$. Because $\pi_1(M)$ is finite, some power of α is null-homotopic. Cap this power by a disk and perturb the capping map, fixed near the boundary, to a Morse function into $(M, \mathcal{F})$. The induced singular foliation of the disk has centres and saddles, and the number of centres is one larger than the number of saddles.

Start at a centre. Near it, the concentric circular leaves map to curves contractible in their leaves. Moving outward, if the first circular leaf which fails to be contractible in its own leaf occurs before a saddle, it is a vanishing cycle. If the boundary of the concentric family has no singularity, then it has non-trivial one-sided holonomy; since the circles just inside it are contractible in their leaves, this boundary curve is again a vanishing cycle. If every boundary curve contains a saddle, then, as in the proof of Haefliger's theorem, two centres share a saddle. If no vanishing cycle has yet appeared, the corresponding boundary curves are contractible in their common leaf; the composite curve has trivial holonomy, so we can pass to concentric circles outside it and continue. This process cannot continue indefinitely because the disk has finitely many singularities. Hence a vanishing cycle exists.

Step 2: a simple vanishing cycle produces disks in nearby leaves. Replace the vanishing cycle by a simple positive one and call it $\alpha_0 \subset L_0$. For small $t > 0$, the curve α_t bounds a disk in L_t after lifting to the universal covering space. More precisely, there is an immersion $A : D \times (0,\varepsilon) \longrightarrow M$ such that $A_t = A(-,t) : D \to L_t$ extends α_t, and A_t lifts to an embedding $\widetilde{A}_t : D \longrightarrow \widetilde{L}_t$. Write $D_t = A_t(D)$.

Because α_0 is not contractible in L_0, this family of disks cannot extend continuously to $t = 0$. Compactness of M then gives a leaf $L \neq L_0$ and a decreasing sequence $t_m \to 0$ such that $L_{t_m} = L$ and $D_{t_m} \subset \mathrm{Int}(D_{t_{m+1}})$. On the universal cover of L, the lifted disks are nested and exhaust $\widetilde{L}$.

Step 3: the leaf containing the vanishing cycle is compact. Suppose, for contradiction, that L_0 is non-compact. We know that every point of a non-compact leaf of a codimension 1 transversely oriented foliation on a compact 3-manifold lies on a closed curve transverse to the foliation. Choose such a closed transverse curve γ through a point of α_0, arranged so that initially it follows the positive transverse family α_t and otherwise avoids the curves α_t.

For a fixed m, the restriction $A : D \times [t_{m+1}, t_m] \longrightarrow M$ factors through a quotient space Y obtained by identifying the disk at t_m with its image inside the disk at t_{m+1}. The induced map $f : Y \longrightarrow M$ is a proper immersion, hence has finite fibres. The transverse curve γ lifts through Y. The orientation of γ and the avoidance condition keep this lift from meeting the boundary of Y, so it can be continued forever. This gives an infinite non-periodic lift over the same closed curve, contradicting the finite fibres of f. Therefore L_0 is compact. □

References

[1] Ieke Moerdijk and Janez Mrčun, *Introduction to Foliations and Lie Groupoids*, Cambridge Studies in Advanced Mathematics, vol. 91, Cambridge University Press, Cambridge, 2003.